

From hyperspectral CHRIS to four RGBN bands

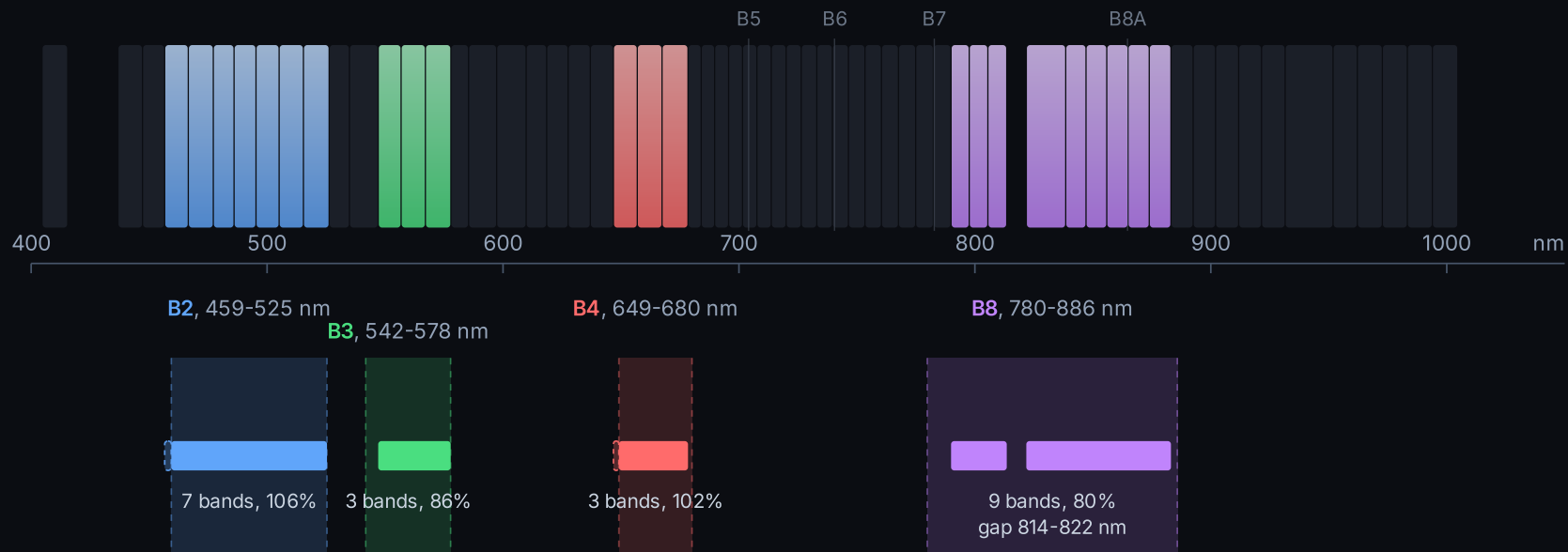
The cloud model eats **4 bands** (R, G, B, NIR). CHRIS has **18 to 62 bands** per mode. So each pseudo-band is built from the CHRIS bands that fall in it.

which each CHRIS band goes to the Sentinel-2 band it overlaps the most (max spectral overlap)

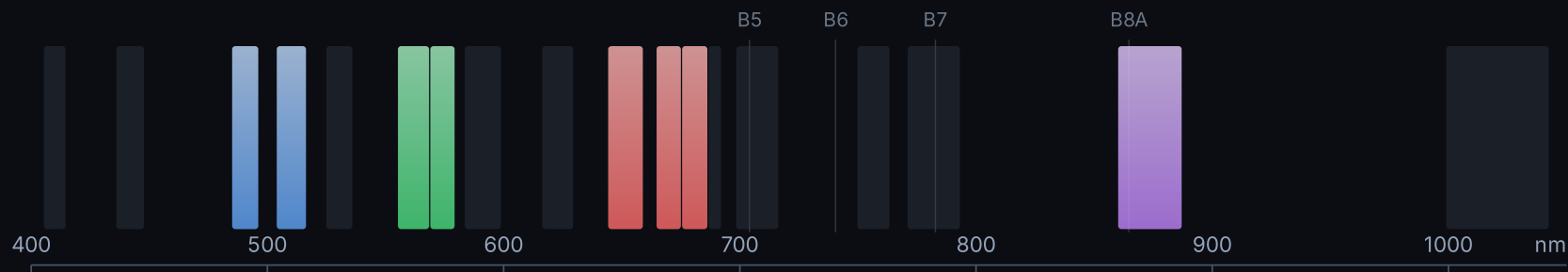
how plain **mean** of the CHRIS bands assigned to each pseudo-band

then DN/100 000 or TOA/10 000, clip to 5, and a per-tile z-score inside the network

Mode 1 : 62 bands, ~34 m



Mode 2 : 18 bands, ~17 m

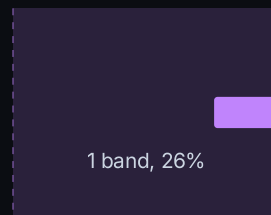
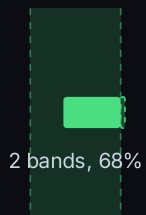
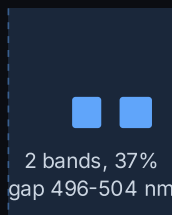


B2, 459-525 nm

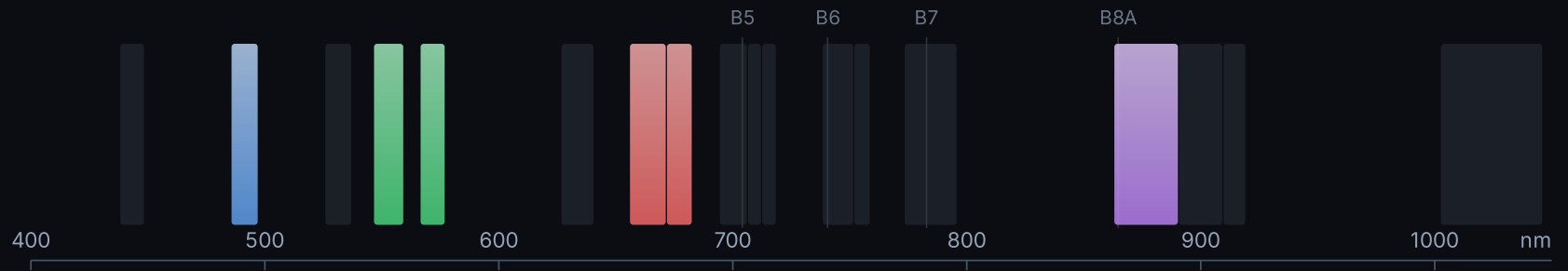
B3, 542-578 nm

B4, 649-680 nm

B8, 780-886 nm



Mode 3 : 18 bands, ~13 m

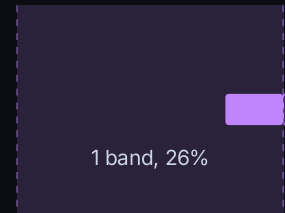
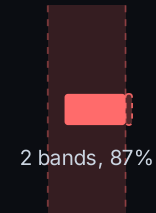


B2, 459-525 nm

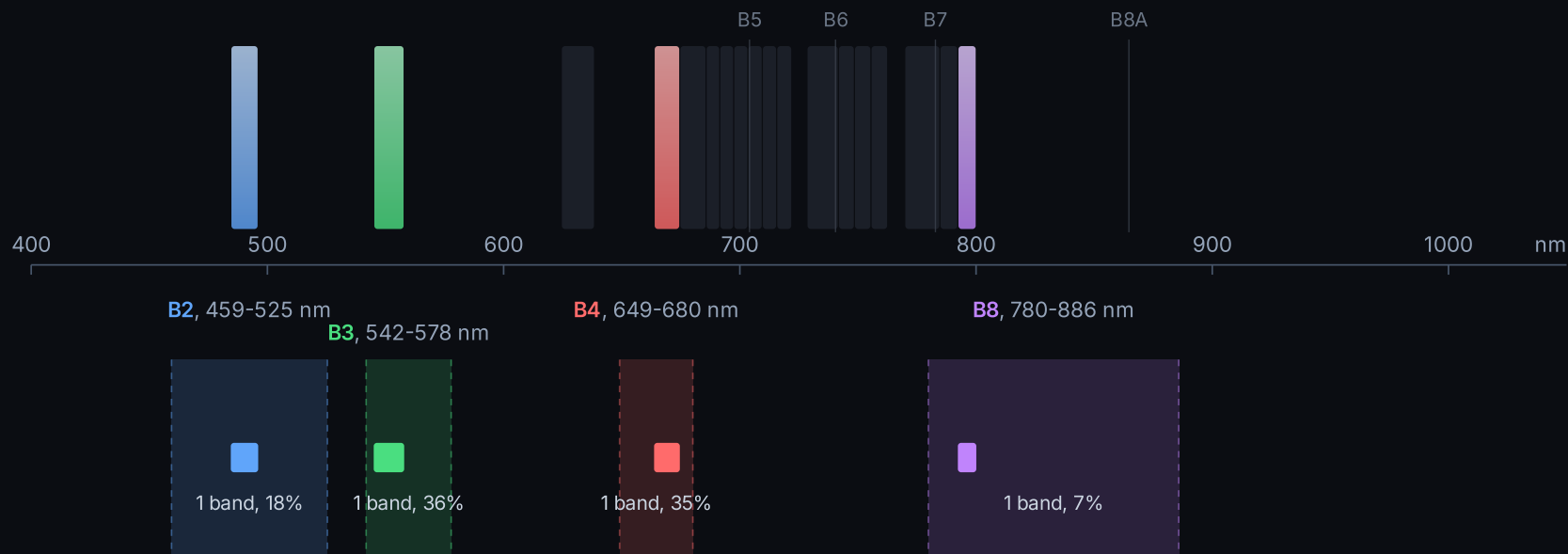
B3, 542-578 nm

B4, 649-680 nm

B8, 780-886 nm

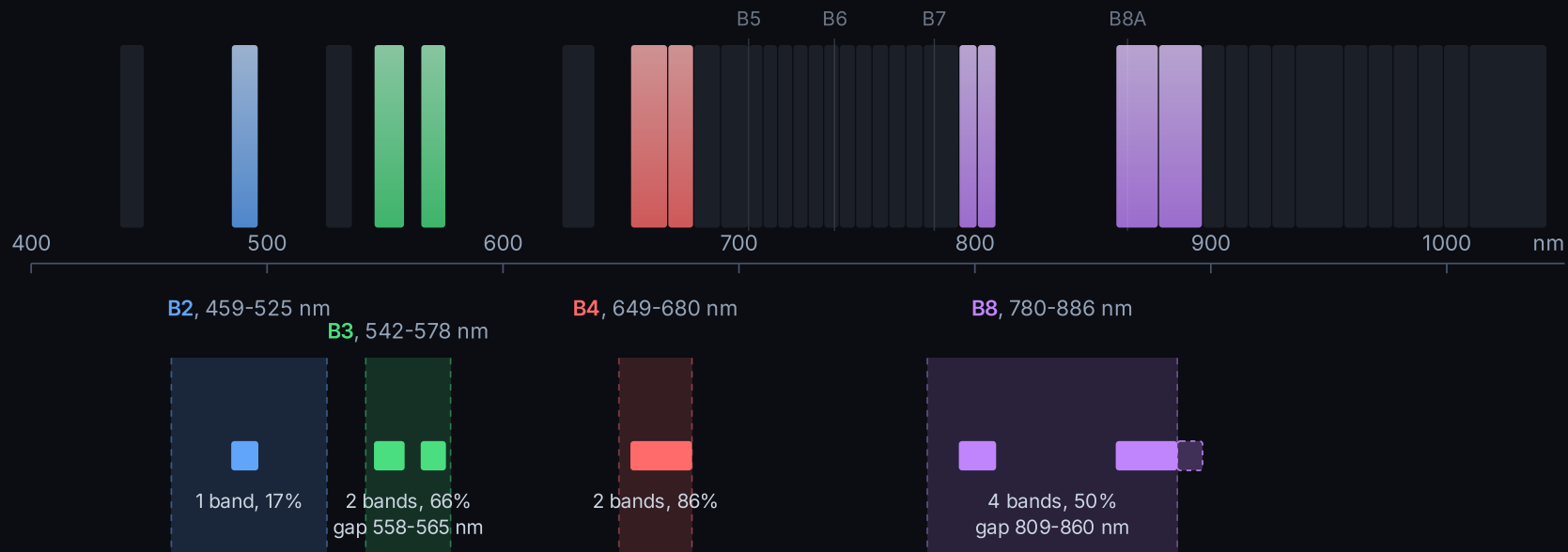


Mode 4 : 18 bands, ~11 m

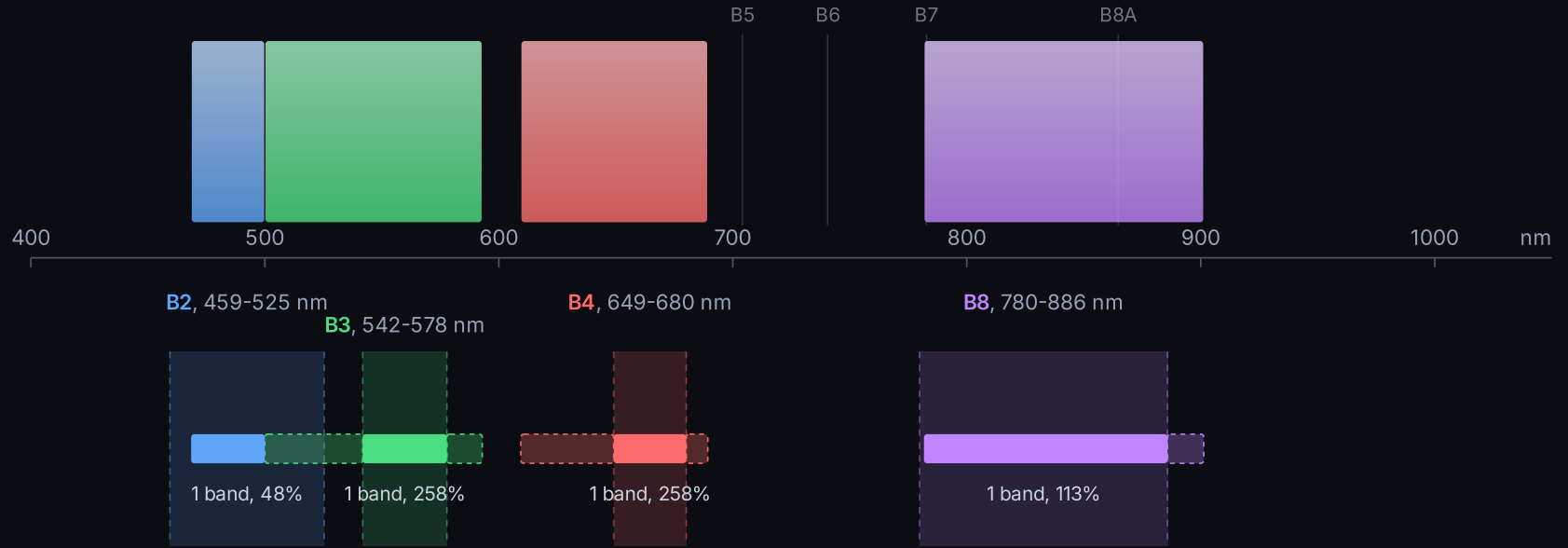


Mode 4 has one clean band per target: 1 → B, 2 → G, 4 → R, 18 → NIR.

Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Mode 20 has exactly 4 bands: no averaging. Each one is wide and sits shifted from the S2 range it falls into.

Where the bands end up

R, B4

G, B3

B, B2

NIR, B8

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	4-10 (7), 457-526 nm	13-15 (3), 547-578	23-25 (3), 647-678	43-51 (9), 790-883
2	3-4 (2), 485-517	6-7 (2), 555-580	10-12 (3), 644-686	17 (1), 860-887
3	2 (1), 486-497	4-5 (2), 546-577	7-8 (2), 656-683	15 (1), 863-890
4	1 (1), 484-496	2 (1), 545-558	4 (1), 664-675	18 (1), 792-800
5	2 (1), 485-496	4-5 (2), 545-576	7-8 (2), 654-681	23-26 (4), 793-896
20	1	2	3	4

Numbers are 1-based CHRIS band indices; nm is the spectral span. Stack order for the model: R, G, B, NIR.